

Lucas Lyons

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Education

University of Illinois, Urbana-Champaign (Urbana-Champaign, IL)

Aug 2026 - Expected Dec 2027

MS, Statistics

- Expected coursework: Statistical inference, regression analysis, deep learning, big data analytics, time series analysis, Bayesian analysis, risk modelling.

University of Toronto, St. George (Toronto, ON)

Jun 2024

Honours BSc, Mathematics and Statistics | Major GPA 3.78/4.00

- Coursework: calculus, real analysis, linear algebra, ODEs, linear and nonlinear optimization, probability, regression analysis, data analysis, complex analysis, group theory, logic.
- Graduate-level coursework: stochastic processes, statistical learning, high-dimensional statistics.

Work Experience

Technology Consulting Staff | EY Canada (Toronto, ON)

Oct 2024 - Mar 2026

- **QA Developer:** Developed modular Python (pandas) and SQL scripts for automated validation of Azure Data Factory (ADF) pipelines across Azure SQL Database and Blob Storage.
- **Project Management:** As the sole EY representative on the client team, managed multiple workstreams supporting the multimillion-dollar IT infrastructure migration of 43 bank branches. Delivered 100% of critical objectives on schedule. Coordinated vendors, managed deadlines, and escalated risks, reporting to the client's senior leadership.

Research and Technical Experience

Languages: Python, SQL, R, JavaScript/TypeScript

Tools: Git, Azure Database, PostgreSQL (Supabase), Excel

Libraries: pandas, scikit-learn, NumPy, matplotlib, Optuna, scrapy, pgvector

Brewtiful (End-to-end Recommendation System) ❖ [App](#) ❖ [Blog](#)

Aug 2025 - Nov 2025

- Built and deployed an end-to-end recommendation platform over 2.5M scraped interactions and 60K+ beers, using LightFM embeddings, PostgreSQL/pgvector with HNSW indexing, and a weighted k-means reranker.
- Developed and tuned LightFM models on a separate 1.5M-review benchmark using temporal validation; achieved 0.902 AUC and 0.174 Precision@50 (approximately 2.4× the popularity baseline). Incorporating metadata features increased catalog coverage at K=10 from 4% to 10%.

Research Assistant, Statistics | (University of Toronto)

Jun 2024 - Aug 2024

Supervised by Professor Piotr Zwiernik, conducted a literature review on the topics of E-values, anytime-valid inference, and sequential testing. Analyzed papers by leading researchers on these subjects including A. Ramdas and P. Grunwald. Reinforced knowledge on topics such as hypothesis testing and exponential family distributions.